BIBEK POUDEL

Collinsville, IL

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EDUCATION

Southern Illinois University Edwardsville (SIUE) — Edwardsville, IL
Master of Science in Geography — May 2026

Tribhuvan University — Nepal
Bachelor of Engineering — July 2022
Specialty: Geomatics Engineering — Percentage: 71.62%

PROFESSIONAL EXPERIENCE

Climate Fellow — Jun 2022- Nov 2022
Youth Innovation Lab — Rautahat, Nepal

- Assessed the DRRM and Climate Change-related policies of Garuda Municipality
- Sensitized municipal staff on DRRM and Climate Change through introductory presentations and on-the-job training
- Mapped critical infrastructures through OSM and data collection, and integrated the collected data into the BIPAD Portal
- Mapped disaster-prone areas within the municipality using PGIS
- Performed Vulnerability and Capacity Assessment (VCA) of each ward of the municipality in close coordination with municipal and ward representatives
- Assisted in the preparation of a draft LDCRP by finding disaster events from the last 30 years and preparing plans with consultation from local people and ward representatives.

Geomatics Instructor — Jan 2023 - Feb 2024
Shree Amarjyoti Secondary School — Birendranagar, Nepal

- Gave theoretical and practical classes on courses (GIS Basics, GIS Applications, Remote Sensing, Photogrammetry, Control Survey, Land Administration and Land Law, Cartography, and Geodesy and Astronomy) to the students of the Diploma in Geomatics Engineering
- Taught how to use advanced surveying instruments like Total Station, Theodolite, and Leveling machines,
- Supervised students in survey camps where they performed traverse, trilateration, triangulation, and leveling surveys.

Research Assistant — Feb 2024- July 2024
Institute of Forestry, Pokhara Campus — Pokhara, Nepal

- Validated previous research on Abandoned Cropland Mapping through a literature review
- Developed a robust methodology for assessing several AF options on abandoned land using open-access spatiotemporal earth observation data in the GEE cloud computing platform.
- Developed a decision support system in Google Earth Engine, where the user can view the abandoned land at the municipal level within the Gandaki Province, along with the suitability for Agroforestry in that abandoned land. (Decision Support System for Agroforestry Suitability in Gandaki Province)

Research Assistant May 2025- Present
Geospatial Mapping, Applications, and Research Center (GeoMARC) Edwardsville, IL

- Serving as a surveyor for the Subsidence Monitoring Response Team, where I actively contribute to the weekly monitoring and analysis of subsidence in various locations that were previously utilized for mining activities.
- My responsibilities involve conducting precise leveling surveys along with team members to assess ground stability and measure potential land subsidence
- Mapping mine subsidence throughout the Madison and St. Clair County using State LiDAR point Cloud

SKILLS

GIS Software (QGIS, ArcGIS Pro, ERDAS Imagine)
Google Earth Engine
OpenStreetMap
Microsoft Office
GPS
Machine Learning
Deep Learning
Programming (R, Python, JavaScript)
Surveying (Total Station, Theodolite, Digital Leveling Machine)
WebGIS
Animated Cartography
LiDAR

TRAINING AND SEMINARS

Disaster Risk Management Localization Manual	Aug 2022
Intermediate Machine Learning	Dec 2022
Pandas	Dec 2022
Python	Dec 2022
Feature Engineering	Dec 2022
Data Visualization	Dec 2022
Data Cleaning	Jan 2023
Use of a Mobile-based Data Collection Tool to Collect Data on Community Resources & Transport Infrastructures for Mobility Planning	March 2023
Synergizing Disaster Risk Reduction and Climate Change Adaptation - Thought Leadership Course	Aug 2023
Google Data Analytics Specialization	Sept 2023
Please Delete Your Shapefiles: Teaching Web-Based Modern GIS	Sept 2023
Spatial Data Science: The New Frontier in Analytics	Oct 2023
EO&GEO Series: Drones for Rural Development	Oct 2023
Imagery in Action	Oct 2023
GIS for Climate Action	Dec 2023
Big Data Analysis for Water-related Applications	Jan 2024
Going Places with Spatial Analysis	Mar 2024
CCAI Virtual Summer School 2024	Jun-Aug 2024